

AMENDMENT NO. 2 JULY 2018
TO
IS 12640 (PART 2) : 2016/IEC 61009-1 : 2012 RESIDUAL CURRENT OPERATED
CIRCUIT-BREAKERS WITH INTEGRAL OVERCURRENT PROTECTION FOR
HOUSEHOLD AND SIMILAR USES (RCBOs)

PART 2 GENERAL RULES

(Second Revision)

This amendment no. 2 is identical to Amendment 2 to IEC 61009-1 : 2012 issued by International Electrotechnical Commission. The contents of the Corrigendum 1 of January 2014, issued by International Electrotechnical Commission, have been included in this copy.

8.1.3 Clearances and creepage distances (see also Annex B)

Replace the third paragraph of this subclause, and the modifications brought to it by Amendment 1, as follows:

The clearances of items 2 and 4 (except accessible surface after installation, see Note 1) may be reduced provided that the measured clearances are not shorter than the minimum allowed in IEC 60664-1 for homogenous field conditions.

Add the following new note after the third paragraph and renumber the existing notes of this subclause, introduced by Amendment 1, accordingly:

NOTE 1 Accessible surface after installation means any surface accessible by the user when the RCD is installed according to the manufacturer's instructions. The test finger can be applied to determine whether a surface is accessible or not.

8.5 Operating characteristics

Delete "and 9.21" from the existing text

8.15 Behaviour of RCBOs in case of earth fault currents comprising a d.c. component

Replace the second paragraph as follows:

Compliance is checked by the tests of 9.9.3.

Table 12 – List of type tests

Delete the following item:

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- Behaviour of RCBOs in case of an earth fault current comprising a d.c. component
9.21

9.7.2 Insulation resistance of the main circuit

Replace the existing text in item c) with the following text:

- c) with the RCBO in the closed position, between all poles connected together and the frame including a metal foil or part in contact with the outer surface of the housing of insulating material but with the terminal areas kept completely free to avoid flashover between terminals and the metal foil;

Replace 9.9.1 and the modifications brought to it by Amendment 1 by the following:

9.9.1 Verification of the operating characteristics under residual current conditions

9.9.1.1 Test circuit and test procedure

The RCBO is installed as for normal use.

The test circuit shall be of negligible inductance. For tests according to 9.9.1.2, the test circuit shall correspond to Figure 4. For tests according to 9.9.1.3, the test circuit shall correspond to Figure 5 or Figure 6, as applicable.

The instruments for the measurement of the residual current shall display (or allow to determine) the true r.m.s. value.

NOTE The information for instrument measurement is available at the following CTL webserver:

[<http://www.iecee.org/ctl/sheet/pdf/CTL%20DSH%20251B%20Beijing%202009_05_15.pdf>](http://www.iecee.org/ctl/sheet/pdf/CTL%20DSH%20251B%20Beijing%202009_05_15.pdf)

Unless otherwise specified, the tests are performed with no load at the reference temperature of $20\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$.

The RCBO shall perform the tests of 9.9.1.2, 9.9.1.3 and 9.9.1.4, as applicable. Each test is made on one pole only, taken at random, with five measurements, unless otherwise specified.

For RCBOs having more than one rated frequency, the tests shall be carried out at the lowest and highest frequency, except for test in 9.9.1.2 e), where verification is performed at only one frequency.

For RCBOs having multiple settings of residual operating current, the tests shall be made for each setting.

9.9.1.2 Tests for all RCBOs

The test conditions according to 9.9.1.1 apply to all RCBOs.

- a) Verification of correct operation in case of a steady increase of the residual current.

The test switches S_1 and S_2 and the RCBO being in the closed position, the residual current is steadily increased, starting from a value not higher than $0,2 I_{\Delta n}$, trying to attain the value of $I_{\Delta n}$ within 30 s, the tripping current being measured each time.

All five measured values shall be situated between $I_{\Delta n0}$ and $I_{\Delta n}$.

- b) Verification of correct operation on closing on a residual current

The test circuit being calibrated at the value of the rated residual operating current $I_{\Delta n}$ and the test switches S_1 and S_2 being closed, the RCBO is closed so as to simulate service conditions as closely as possible. The break time is measured five times. No measurement shall exceed the limiting value specified for $I_{\Delta n}$ in Table 2, according to the type of RCBO.

- c) Verification of correct operation in case of sudden appearance of residual current

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1) All types

The test circuit being successively calibrated at each of the values of residual current specified in Table 2, the test switch S_2 and the RCBO being in the closed position, the test voltage is suddenly established by closing the test switch S_1 .

The RCBO shall trip during each test.

Five measurements of the break time are made at each value of residual current.

No value shall exceed the relevant specified limiting value given in Table 2.

2) Additional test for type S

The test circuit being successively calibrated at each of the values of residual current specified in Table 2, the test switch S_1 and the RCBO being in the closed position, the residual current is suddenly established by closing the test switch S_2 for periods corresponding to the relevant minimum non-actuating times, with a tolerance of 0_{-5}^0 %.

Each application of residual current shall be separated from the previous one by an interval of at least 1 min.

The RCBO shall not trip during any of the tests.

d) Verification of correct operation in case of sudden appearance of residual currents between $5 I_{\Delta n}$ and 500 A

The test circuit is calibrated at any two values of the residual current chosen at random within the range 5 A to 200 A, among the following list 5 A, 10 A, 20 A, 50 A, 100 A, 200 A.

NOTE In Australia the measurement of the break time is made at 5 A, 10 A, 20 A, 50 A, 100 A, 200 A.

The test switch S_1 and the RCBO being in the closed position, the residual current is suddenly established by closing the test switch S_2 .

The RCBO shall trip during each test. The break time shall not exceed the times given in Table 2.

The test is made once for each value of residual current on one pole only, taken at random.

e) Verification of correct operation with load

The tests of 9.9.1.2 b) and 9.9.1.2 c) are repeated, the pole under test and one other pole of the RCCB being loaded with rated current, this current being established shortly before the test.

For the test of 9.9.1.2 c), the switch S_1 and the RCBO are in closed position. The residual current is established by closing S_2 .

f) Tests at the temperature limits

The RCBO shall perform the tests specified in 9.9.1.2 c) under the following conditions, successively:

1) ambient temperature: -5°C , off-load;

2) ambient temperature: $+40^{\circ}\text{C}$, the RCBO having been previously loaded with the rated current, at any convenient voltage, until it attains thermal steady-state conditions.

In practice these conditions are reached when the variation of temperature-rise does not exceed 1 K per hour.

For the tripping tests in 2), the flow of rated current may be interrupted, provided that the total interruption period does not exceed 30 s. As soon as the sum of interruption periods exceed 30 s, the RCBO shall be loaded again with rated current for 5 min before next tripping time measurement.

NOTE Preheating may be carried out at any convenient voltage at either 50 Hz or 60 Hz but auxiliary circuits shall be connected to their normal operating voltage (particularly for components depending on line voltage).

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9.9.1.3 Verification of correct operation at residual currents with d.c. components

This subclause applies to type A RCBOs, under test conditions of 9.9.1.1.

- a) Verification of correct operation in case of a continuous rise of residual pulsating direct current

The test shall be performed according to Figure 5.

The auxiliary switches S_1 and S_2 and the RCBO shall be closed. The relevant thyristor shall be controlled in such a manner that current delay angles α of 0° , 90° and 135° are obtained. Each pole of the RCBO shall be tested twice at each of the current delay angles, in position I as well as in position II of the auxiliary switch S_3 .

For each test, the current shall be steadily increased at an approximate rate of $1,4 I_{\Delta n} / 30$ amperes per second for RCBOs with $I_{\Delta n} > 0,01$ A, and at an approximate rate of $2 I_{\Delta n} / 30$ amperes per second for RCBOs with $I_{\Delta n} \leq 0,01$ A, starting from zero. The tripping current shall be in accordance with Table 26.

- b) Verification of correct operation in case of suddenly appearing residual pulsating direct currents

The RCBO shall be tested according to Figure 5.

The circuit being successively calibrated at the values specified hereafter and the auxiliary switch S_1 and the RCBO being in the closed position, the residual current is suddenly established by closing the switch S_2 .

The test is carried out at each value of residual current specified in Table 3, according to the type of RCBO.

Two measurements of the break time are made at each value of residual current, at a current delay angle $\alpha = 0^\circ$ with the auxiliary switch S_3 in position I for the first measurement and in position II for the second measurement.

No value shall exceed the specified limiting values.

- c) Verification of correct operation with load

The tests of 9.9.1.3 a) are repeated, the pole under test and one other pole of the RCBO being loaded with the rated current, this current being established shortly before the test.

NOTE The loading with rated current is not shown in Figure 5.

- d) Verification of correct operation in case of residual pulsating direct currents superimposed by a smooth direct current of 0,006 A

The RCBO shall be tested according to Figure 6 with a half-wave rectified residual current (current delay angle $\alpha = 0^\circ$) superimposed by a smooth direct current of 0,006 A.

Each pole of the RCBO is tested in turn, twice at each of positions I and II.

The half-wave current I_1 , starting from zero, being steadily increased at an approximate rate of $1,4 I_{\Delta n} / 30$ amperes per second for RCBOs with $I_{\Delta n} > 0,01$ A and $2 I_{\Delta n} / 30$ amperes per second for RCBOs with $I_{\Delta n} \leq 0,01$ A, the device shall trip before this half-wave current I_1 reaches a value not exceeding $1,4 I_{\Delta n}$ or $2 I_{\Delta n}$ respectively.

Table 26 – Tripping current ranges for type A RCBOs

Angle, α °	Tripping current A	
	Lower limit $I_{\Delta n}$	Upper limit $I_{\Delta n}$
0	0,35	} 1,4 or 2 (5.3.8)
90	0,25	
135	0,11	

9.9.1.4 Particular test conditions for RCBOs functionally dependent on line voltage

For RCBOs functionally dependent on line voltage, each test is made at each of the following values of the line voltage, applied to the relevant terminals: 1,1 and 0,85 times the rated line voltage.

9.12.11.2.1 Tests on all RCBOs

Replace the second paragraph of this subclause, and the modifications brought to it by Amendment 1, by the following:

Each overcurrent protected pole of the RCBO is subjected separately to a test in a circuit similar to Figure 7, but with the impedance Z1 connected only to the pole to be tested and directly connected to the Neutral without passing the current through the N-pole of the RCBO, at a voltage of 105% of the rated phase to neutral voltage.

9.21 Verification of correct operation of residual currents with d.c. components

Replace the title and text of this subclause by "Void".

Replace 9.24 by the following:

9.24 Electromagnetic compatibility (EMC)

9.24.1 Tests covered by the present standard

Tests listed in Table 27 are covered by the present standard and need not to be repeated.

Table 27 – Tests covered by this standard

Reference to Tables 4 and 5 of IEC 61543: 1995 and Amendment 1: 2004	Electromagnetic phenomena	Tests of IEC 61009-1
T.1.3	Voltage amplitude variations	9.9.1.4 and 9.17
T.1.4	Voltage unbalance	9.9.1.4 and 9.17
T.1.5	Power-frequency variations	9.2
T.1.8	Magnetic fields	9.12
T.2.4	Current oscillatory transients	9.19

9.24.2 Additional tests

Tests listed in Table 29 shall be carried out according to the test sequences H, I and J listed in Annex A of the present standard.

Table 29 – Test to be carried out according to IEC 61543

Reference to Tables 4, 5 and 6 of IEC 61543: 1995 and Amendment 1: 2004	Electromagnetic phenomena
T1.1	Harmonics, interharmonics
T1.2	Signalling voltage
T2.3	Surges
T2.1	Conducted sine-wave form voltages or currents
T2.5	Radiated electromagnetic field
T2.2	Fast transients (burst)

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T2.6	Conducted common mode disturbances in the frequency range lower than 150 kHz
T3.1	Electrostatic discharges

For devices containing a continuously operating oscillator, the test of CISPR 14-1 shall be carried out on the samples prior to the tests of IEC 61543.

Figure 26 – Damped oscillator current wave, 0,5 μ s/100 kHz

Add the following note to the figure:

NOTE Care should be taken that the oscillating wave is guaranteed at least up to the 5th full period (50 μ s).

Annex A – Test sequence and number of samples to be submitted for certification purposes

Table A.1 – Test sequences

Delete "withstanding high d.c. voltages due to insulation measurements" from row B 9.7.6.

Delete "9.21 D.C. components" from row D₁.

Add a footnote ^b to test sequence H:

^b For devices containing a continuously operating oscillator, the test of CISPR 14-1 shall be carried out on the samples prior to the tests of this sequence.

Table A.2 – Number of samples for full test procedure

Replace the table header in the third column by "Minimum number of accepted samples".

Delete "Maximum" from the table header in the fourth column.

Table A.3 – Number of samples for simplified test procedure

Add a footnote ^k to test sequence H:

^k If the requirement to test max. rating I_n and minimum rating $I_{\Delta n}$ does not cover all the possible range of RCBOs, the minimum $I_{\Delta n}$ shall in any case be chosen for the test."

Annex IE – Follow-up testing programme for RCBOs

Table IE.1 – Test sequences during follow-up inspections

Replace "9.9.1.4" by "9.9.1.2 f)" in sequence Y1.